

Effects of radiant temperature on thermal comfort

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Abstract

The aim of this paper is to investigate the local differences between body segments caused by high radiant temperature, and to analyze the interior surface temperatures for different wall and ceiling constructions with their effect on thermal comfort. For the segment-wise thermal interactions between human body and its surrounding, simulations have been conducted by appropriately modifying Gagge 2-node model to multi-segment case to demonstrate the local differences. Simulation results are found to be in good agreement with experimental and simulation results reported in the literature. To calculate the interior surface temperatures of the wall and ceiling, the sol-air temperature approach is used for convenience. It is shown in the paper that the body segments close the relatively hot surfaces are more affected than others and interior surface temperatures of un-insulated walls and ceilings exposed to a strong solar radiation reach high levels, all of which cause thermal discomfort for the occupants in buildings.

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1. Introduction

The good building design characteristics, including both the engineering and non engineering disciplines, might be summarized as follows [1]:

- meets the purpose and needs of the building's owners/managers and occupants,
- meets the requirements of health, safety and environmental impact as prescribed by codes and recommended by consensus standards,
- achieves good indoor environment quality which in turn encompasses high quality in the following dimensions: *thermal comfort*, indoor air quality, acoustical comfort, visual comfort,
- creates the intended emotional impact on the building's occupants and beholders.

Thermal comfort is defined as “that condition of the mind in which satisfaction is expressed with the thermal

environment” [2]. Six primary factors identified to most affect the thermal comfort are air temperature, humidity, air velocity, *mean radiant temperature*, clothing level and metabolic rate. The mean radiant temperature is a significant factor, especially in buildings whose envelopes were exposed to a strong solar radiation, that conventional indoor temperature and humidity control cannot guarantee indoor comfort. Cold walls or windows may cause a person to feel cold even though the surrounding air may be at a comfortable level. Likewise warm surfaces may cause a person to feel warmer than the surrounding air temperature would indicate. The building envelopes are the main factors of building energy efficiency and human thermal comfort, as they represent a skin of the building's body. The building whose envelopes include suitable insulation has little internal heat gains and outside gains from solar radiation, and in such a situation, the interior surface temperature of the building walls helps in defending from outside environmental conditions. In buildings where envelopes are un-insulated, the interior surface temperature of the building walls is affected from the outside environmental conditions, especially the solar radiation. Therefore, the interior surface temperatures of these walls rapidly increase.

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Nomenclature

A	surface area, m ²
C	convective heat loss from skin, W/m ²
c_p	constant pressure specific heat, kJ/kgK
E	evaporative heat loss, W/m ²
F	view factor, dimensionless
h	heat transfer coefficient, W/m ² K
i	segment number
I	solar radiation, W/m ²
j	surface number
L	thermal load (W/m ²)
m	body segment mass, kg
M	rate of metabolic heat production, W/m ²
PMV	predicted mean vote index, dimensionless
q	heat flow, W
Q	heat flow rate, W/m ²
RH	relative humidity, dimensionless
S	heat storage, W/m ²
T	temperature, °C
W	external work (W/m ²)
α	absorptance of the surface and fraction of body mass concentrated in skin compartment, dimensionless

ΔR	radiation exchange, W/m ²
ε	emittance of the surface, dimensionless
θ	time, s
$\sum R_i$	thermal resistance of the wall or ceiling excluding the outside air film resistance, m ² K/W
$\sum R_{ii}$	thermal resistance of the wall or ceiling excluding the inside and outside air film resistances, m ² K/W

Subscripts

a	air or ambient
cr	core
cr,sk	from core to skin
e	sol–air
int	interior surface
N	surface number
o	outside
p	person
r	radiation
res	respiration
s	exterior surface
sk	skin

Recently, several studies dealing with thermal comfort, especially radiant temperature have been carried out by researchers. Detailed radiation properties for a thermal manikin were predicted numerically by Sorensen [3]. In the study, the view factors between individual body segments and between the body segments and the outer surfaces were determined and it was pointed out that radiation between individual body segments is important. Similarly, Sakoi et al. [4] proposed a method for estimating the view factor that can be used in place of diagrams for large- and medium-scale spaces. Experimental study dealing with radiant temperature effects on thermal comfort with Predicted Mean Vote (PMV) scale was performed by Yang and Su [5]. Olesen and Parsons [6] described existing ISO standards and current activity concerned with thermal comfort including draught, vertical air temperature difference, floor temperature and radiant asymmetry which cause mostly local thermal discomfort. Holz et al. [7] dealt with energy performance simulation programme accompanied by comfort indications such as mean radiant temperature and air velocity. Butera [8] described the factors which affect thermal comfort conditions in detail and also summarized heat and mass transfer equations between body and environment. In addition, the change of PMV indices with operative temperature at various activity levels, air velocity and clothing were given. Kaynakli et al. [9] investigated the human thermal comfort for sitting and standing posture under steady-state conditions and Kaynakli and Kilic [10] examined the effects of clothing and air velocity on thermal comfort under transient conditions.

But in these studies, the mean radiant temperature was assumed to be equal with ambient air temperature. In order to improve the present understanding of the human thermal comfort and HVAC system, further investigations still have to be carried out. On the other hand, Yumrutas et al. [11] developed an analytical solution method for the estimation of space heat gain and interior surface temperatures of buildings, but the effect of surface temperature on human thermal comfort has not been taken into consideration.

In this study, the effects of radiant temperature on the human thermal comfort are investigated. This paper comprises two purposes. One of them is to investigate the local differences between the body segments caused by high radiant temperature. For this aim, thermal interactions between human body and its environment are simulated to predict skin temperature of the individual body segments. In the simulation, PMV index is also calculated to discuss thermal comfort or discomfort. This simulation is based on Gagge 2-node model but includes some significant modifications over the Gagge 2-node model. The simulation is to apply the Gagge 2-node model to individual body segments rather than to the whole body. The human body is described by 16 cylindrical segments representing the head, hands, arms, etc. The required data for each of the body segments such as neutral temperatures of the core and skin layers, surface areas, and weights are taken from the existing literature. Using this simulation, time-dependent sensible and latent heat losses are obtained to determine PMV index. Time-dependent skin temperatures are also

calculated and this value is evaluated at the end of the 1-h period. The simulation results are also compared with available experimental data and simulated results in the literature, in order to validate the presented 16 segments-Gagge 2-node model and it seems that they are in good agreement.

Second aim of this study is to analyze the interior surface temperatures and its effect on human thermal comfort of different wall and ceiling constructions commonly used in Turkey. The sol-air temperature approach is used to calculate the interior surface temperatures of the wall and ceiling because of its convenience. Thermal comfort or discomfort is discussed with PMV index obtained from the simulation mentioned above. So, the effects of radiant temperature on the human thermal comfort in actual environment can be evaluated by application.

2. Calculating the interior surface temperatures of the wall and ceiling

The sol-air temperature approach is used to calculate the interior surface temperatures of the wall and ceiling. For opaque surfaces such as the walls and roof, the effect of solar radiation is conveniently accounted for by considering the outside temperature to be higher by an amount equivalent to the effect of solar radiation. This is done by replacing the ambient temperature in the heat transfer relation through the walls and the roof by the sol-air temperature (T_e), which is defined as the equivalent outdoor air temperature that gives the same rate of heat flow to a surface as would the combination of incident solar radiation (I), convection with the ambient air, and radiation exchange with the sky and the surrounding surfaces ($\varepsilon \Delta R$) [12–14].

Heat flow into an exterior surface of a building subjected to solar radiation can be expressed as:

$$\begin{aligned} q/A &= \alpha I + h_o(T_o - T_s) - \varepsilon \Delta R \\ &= h_o(T_e - T_s). \end{aligned} \quad (1)$$

Eq. (2) expresses the sol-air temperature as a function of the outside dry bulb temperature, (T_o), emittance (ε) and absorptance (α) of the surface, total incident solar radiation, (I), and the convective heat transfer coefficient (h_o) for the surface:

$$T_e = T_o + \alpha I / h_o - \varepsilon \Delta R / h_o. \quad (2)$$

Once the sol-air temperature is available, heat transfer through a wall and hence interior (T_{int}) and exterior (T_s) surface temperatures of the walls can be expressed as

$$q/A = h_o(T_e - T_s) = \frac{T_s - T_a}{\sum R_I} = \frac{T_s - T_{int}}{\sum R_{II}}, \quad (3)$$

$$T_s = \frac{h_o T_e + (T_a / \sum R_I)}{(1 / \sum R_I) + h_o}, \quad (4)$$

$$T_{int} = T_s + h_o \sum R_{II}(T_s - T_e). \quad (5)$$

In these equations, T_a is the inside ambient temperature, $\sum R_I$ the thermal resistance of the wall or ceiling excluding the outside air film resistance and $\sum R_{II}$ the thermal resistance of the wall or ceiling excluding the inside and outside air film resistances.

3. Description of radiant temperature

The radiant temperature can be calculated from measured values of the temperature of the surrounding walls and surfaces and their positions with respect to the person. As most building materials have a high emittance, all the surfaces in the room can be assumed to be black. If relatively small temperature differences exist between the surfaces of the enclosure, the following equation can be used [2]:

$$T_r = T_1 F_{p-1} + T_2 F_{p-2} + \dots + T_N F_{p-N}, \quad (6)$$

where T_N is the temperature of the surface and F_{p-N} the view factor between a person and surface N . For 16-segments model, the radiant temperature for each body segment is written as

$$T_r(i) = \sum_{j=1}^N T(j) F(i, j), \quad (7)$$

where “ i ” and “ j ” represent the segment and surface number, respectively. $F(i, j)$ is the view factor between a body segment of “ i ” and surface “ j ”.

4. Simulation model

In order to demonstrate the local differences between the body parts, the human body is divided into 16 sedentary segments in this computational model. The 16 body parts and their respective surface areas, weights, and neutral core and skin temperatures are listed in Table 1.

In this simulation, Gagge 2-node model is used but it is applied to individual body segments rather than to whole body. In this model, a transient energy balance states that the rate of heat storage equals the net rate of heat gain minus the heat loss and it can be expressed as [2]

$$S_{cr}(i) = M - W - [C_{res}(i) + E_{res}(i)] - Q_{cr,sk}(i), \quad (8)$$

$$S_{sk}(i) = Q_{cr,sk}(i) - [C(i) + R(i) + E_{sk}(i)]. \quad (9)$$

The rate of heat storage in the core and skin layers causes the rate of change in internal energy and so, core and skin temperature changes:

$$S_{cr}(i) = [1 - \alpha(i)]m(i)c_{p,b}[dT_{cr}(i)/d\theta]/A(i), \quad (10)$$

$$S_{sk}(i) = \alpha(i)m(i)c_{p,b}[dT_{sk}(i)/d\theta]/A(i). \quad (11)$$

In these equations, for 16-segments model, the heat loss by respiration ($C_{res} + E_{res}$) is supposed to occur only at the core layer of the chest segment. In addition, heat conduction between body parts and heat transfer between segments via the blood flow are ignored because there are

Table 1

The neutral core and skin temperatures, DuBois surface areas, and weights of the body segments [15]

<i>i</i>	Body segments	Neutral skin temperature (°C)	Neutral core temperature (°C)	Dubois surface area (m ²)	Weight (kg)
1	Left foot	33.9	35.1	0.056	0.480
2	Right foot	33.9	35.1	0.056	0.480
3	Left leg	33.4	35.6	0.112	3.343
4	Right leg	33.4	35.6	0.112	3.343
5	Left thigh	33.8	35.8	0.209	7.013
6	Right thigh	33.8	35.8	0.209	7.013
7	Pelvis	33.4	36.3	0.221	17.57
8	Head	35.6	36.9	0.140	4.020
9	Left hand	35.2	35.4	0.050	0.335
10	Right hand	35.2	35.4	0.050	0.335
11	Left arm	34.6	35.5	0.063	1.373
12	Right arm	34.6	35.5	0.063	1.373
13	Left shoulder	33.4	35.8	0.096	2.163
14	Right shoulder	33.4	35.8	0.096	2.163
15	Chest	33.6	36.5	0.175	12.40
16	Back	33.2	36.5	0.161	11.03
Whole body				1.87	74

very small differences between the temperatures of body segments.

However, sensible (C , R and C_{res}) and latent (E_{res} and E_{sk}) heat loss equations and other required equations for the model are not given in the paper seeing that they are easily reached from the existing literature, such as ASHRAE [2], Butera [8] and Kaynakli et al. [9] and Kaynakli and Kilic [10]. The thermal and evaporative resistances calculation methods for each body segments are taken from Yigit [16] and McCullough et al. [17].

5. Prediction of thermal comfort

In the simulation, PMV index is also calculated to evaluate thermal comfort or discomfort. PMV is an index that predicts the mean value of the votes of a large group of persons on a 7-point thermal sensation scale: 0 *neutral*, ± 1 *slightly warm/cool*, ± 2 *warm/cool*, ± 3 *hot/cold*. The calculation method of the PMV index is presented by ISO 7730 [18] and ASHRAE [2]. Now that, transient energy balance model (or Gagge model) is used in the simulation, the variations dependent on time are taken into consideration. Time-weighted averages of the PMV during the previous 1-h (or 3600 s) period are applied according to ISO 7730 [18].

Firstly thermal load (L) is obtained for each body segment and then total thermal load (L_{total}) on the body is calculated:

$$L(i) = M - W - [C(i) + R(i) + E_{sk}(i) + C_{res}(i) + E_{res}(i)], \quad (12)$$

$$L_{total} = \sum_{i=1}^{16} L(i). \quad (13)$$

After, time-dependent PMV index $[PMV(\theta)]$ is found and time-weighted averages of PMV is calculated for a 1-h period as follows:

$$PMV(\theta) = [0.303 \exp(-0.036M) + 0.028]L_{total}, \quad (14)$$

$$PMV_{total} = \sum_{\theta=1}^{3600} PMV(\theta), \quad (15)$$

$$PMV = PMV_{total}/3600. \quad (16)$$

6. Validation of 16-segments-Gagge 2-node model

In order to present reliability of the results estimated from the 16 segments-Gagge 2-node model mentioned above, the simulation results have been compared with the simulated results and measured data in the literature. Raven and Horvath [19] exposed the subjects to a downward step change from 28 °C at 45% RH to 4.7 °C at 70% RH, in still air. Skin temperatures on segments were presented in that study. Similarly, Hardy and Stolwijk [20] exposed the subjects to a step change from 43 °C at 30% RH to 17 °C at 40% RH. Mean skin temperatures of the subjects for various time steps were given in that study. Stolwijk and Hardy [21] exposed the subjects to upward step change from 30 °C at 40% RH to 48 °C at 30% RH and presented total evaporative heat loss from the subjects. Huizenga et al. [22] presented the simulation results obtained from the Berkeley comfort model based on the Stolwijk model for similar environmental conditions with the mentioned experimental studies, and compared the results with the measured data. In that study, the results estimated from the simulation are compared with the measured and simulated results mentioned above. Measured data given in Raven and Horvath [19], Hardy and

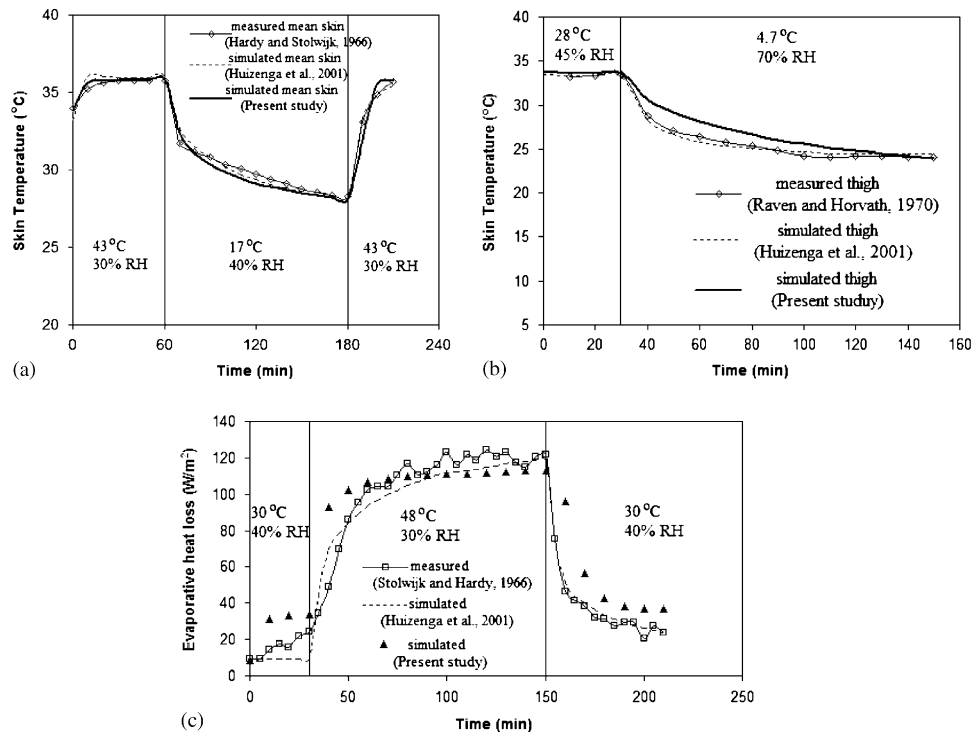


Fig. 1. Comparison of the results predicted by the model with the results reported by other researchers: (a) mean skin temperature, (b) skin temperature on the thigh and (c) evaporative heat loss.

Stolwijk [20] and Stolwijk and Hardy [21] are taken from Huizenga et al. [22]. It can be seen from Fig. 1, that the results predicted from the model presented here are in good agreement with the measured and simulated data taken from the literature mentioned above briefly. Presented model results, simulated results obtained by Huizenga et al. [22] and the measured data in the literature are different from each other in some time steps due to the metabolic activity level and the thermal and evaporative resistances of the clothing differences between our model and the studies in the literature. The predicted temperature and evaporative heat loss agree well with the simulated results and experimental data in the literature, indicating satisfactory predictability of the model.

However, predicted PMV index from the simulation is also compared with ISO 7730 [18]. It can be seen from Fig. 2 that predicted PMV are very close with the values taken from ISO 7730 [18].

7. Investigation of local differences caused by radiant temperature

In this section of the paper, the local differences between the body segments caused by high radiant temperature are investigated with the simulation explained above. In the simulation, the clothing levels were adjusted to hold constant at approximately 0.57 clo for a typical summer clothing (briefs, short-sleeve shirt, long trousers and socks). The activity level was taken as constant at 60 W/m^2 for a seated, quiet person [2]. The air velocity was assumed as 0.2 m/s . Seeing that the influence of humidity on thermal

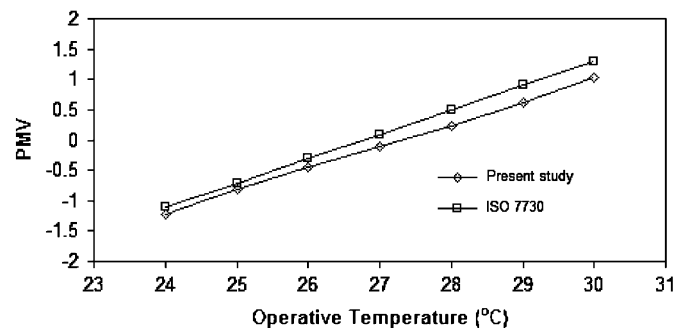


Fig. 2. Comparison of the predicted PMV with ISO 7730 [18] (air velocity: 0.2 m/s , activity level: 60 W/m^2 and clothing resistance: 0.5 clo).

sensation is small at moderate temperatures close to comfort, relative humidity was held constant at 50% for all cases [6,8]. The view factors between the body segments and the outer surfaces given in Table 2 are taken from Sorensen [3]. These values were used in the simulation. The set-up of the Sorensen's [3] study considered the manikin positioned in a room of length, width and height of 2.95, 2.95 and 2.4 m, respectively. The tip of the nose of the manikin was centred between the walls at a height of 1.25 m, whereby the feet were raised 20 mm above the floor. In the mentioned study, the view factors were calculated by the radiation module of the CFD programme.

Due to the fact that there are so many radiant surfaces and objects in a building, determination of the radiant temperature is quite difficult. In order to simplify calculating this temperature, only three radiant surfaces, the two walls and ceiling, are treated in these simulations.

Table 2

View factors $F(i,j)$ (%) used in the simulation [3]. (Because of the symmetrical set-up, only the left side is given)

	Pelvis	Chest	Back	Head	L.foot	L.leg	L.thigh	L.hand	L.arm	L.shoul.
Floor	23.7	11.6	14.2	9.58	51.5	36.9	34.7	31.3	17.6	12.3
Ceiling	8.20	14.3	12.9	20.5	6.01	4.42	10.3	3.54	6.79	11.9
F. W.	11.0	30.9	0.93	12.9	11.2	18.0	10.8	8.78	13.8	14.1
B. W.	24.5	0.87	41.2	15.4	4.65	6.75	4.19	9.81	13.0	17.4
L. W.	10.1	11.9	9.95	16.3	13.3	15.2	15.4	16.4	19.8	23.2
R. W.	9.95	12.6	9.37	16.2	6.89	8.04	5.10	2.34	3.08	1.82
Body	12.6	17.9	11.5	9.16	6.45	10.7	19.6	27.8	25.8	19.4

Front wall: F. W., back wall: B. W., left wall: L. W., right wall: R. W.

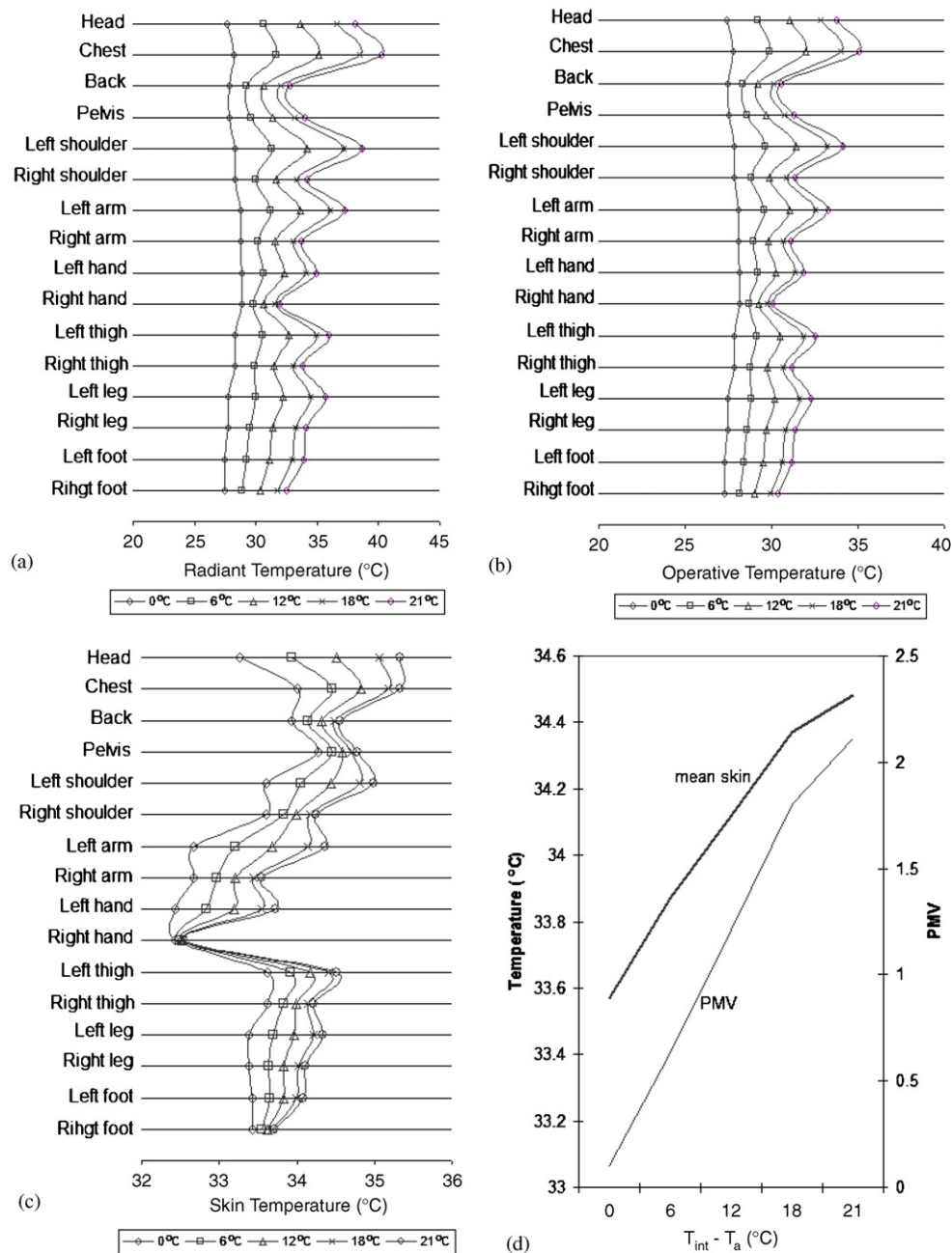


Fig. 3. The effect of temperature differences between interior surfaces and ambient air on the radiant temperatures, operative temperatures, skin temperatures of body parts, mean skin temperature and PMV index depending of body segments.

The other objects and walls are assumed to be in thermal equilibrium with the ambient air temperature. Ambient air temperature is assumed to be constant at 27 °C for all cases, because PMV index equals approximately the neutral value if the operative temperature is 27 °C [18]. But the mean radiant temperature affects the operative temperature and hence the PMV index.

Interior surface temperatures of the ceiling and the left and front walls of the person are assumed as 0, 6, 12, 18 and 21 °C higher than the ambient air temperature (i.e. $T_{\text{int}} - T_a = 0, 6, 12, 18, 21$). The effect of temperature differences between the interior surfaces and the ambient air on the changing of the radiant temperatures, the operative temperatures and the skin temperatures depending on the body segments are shown in Fig. 3 (a)–(c), respectively. It can be seen that an increase in this temperature difference increases the radiant temperatures, effecting the body segments, and hence the operative

temperatures especially in head, left and front side of the body. The left shoulder, chest and head segments are more affected than the other body parts from the ceiling, left and front walls. For that reason, the radiant temperature and the operative temperature of these segments rapidly increase as shown in Fig. 3(a) and (b), respectively. This situation causes local thermal discomfort especially in body parts relatively close to the hot surface. The skin temperatures of the body parts close to the warm surface such as head, chest and left shoulder rapidly increase as seen from Fig. 3(c). Also, it can be seen from Fig. 3(d) that the mean skin temperature rapidly boosts with an increase in the temperature difference.

Although the indoor ambient air temperature and relative humidity have been kept under comfortable level (27 °C and 50% RH in these simulations) for occupants in buildings, the high radiant temperature causes to increase the PMV index. A change in the PMV index according to

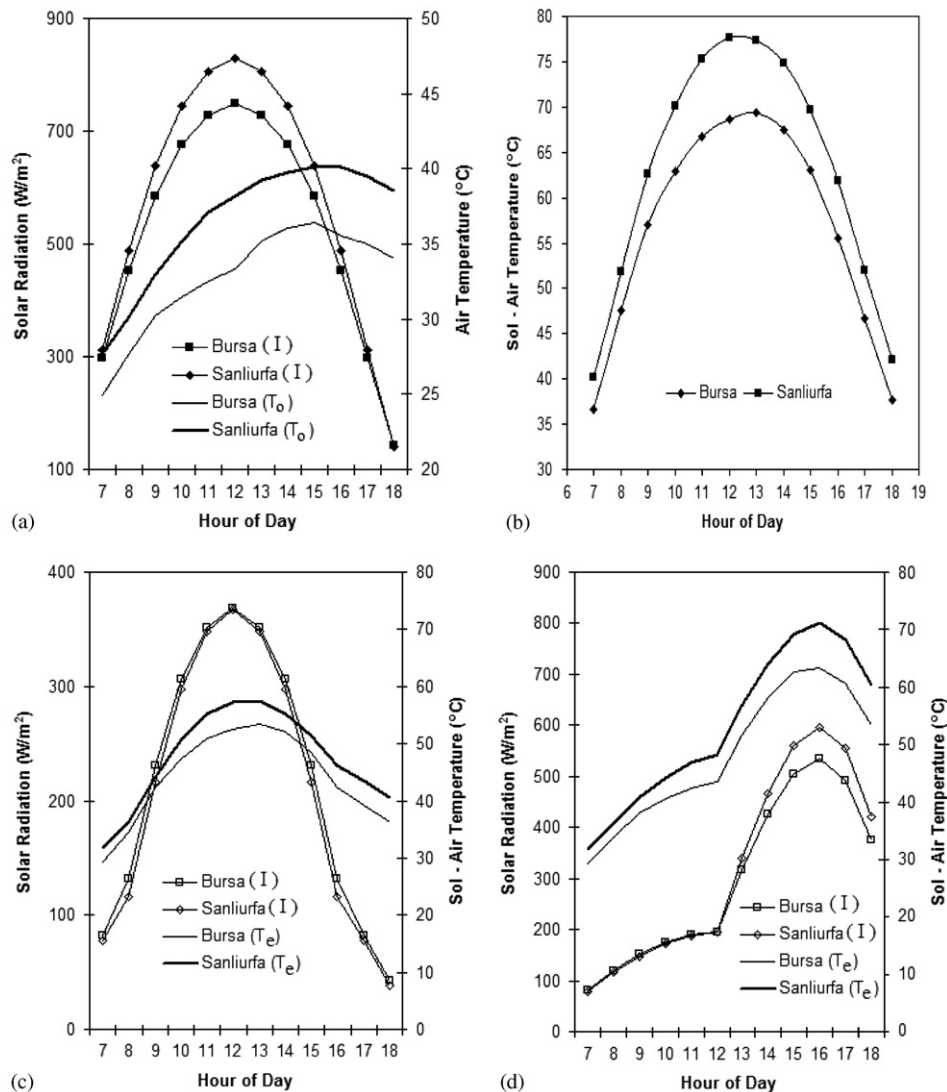


Fig. 4. For the cities of Bursa and Sanliurfa in July: (a) average hourly ambient air temperature and solar radiation on the horizontal surface, (b) the variation of sol-air temperature on the horizontal surface, (c) the variations of solar radiation and sol-air temperature for the south-facing wall and (d) the variations of solar radiation and sol-air temperature for the west-facing wall.

temperature differences between the interior surfaces of ceiling, left and front wall of the occupant and the indoor ambient air ($T_{\text{int}} - T_a$) are shown in Fig. 3(d). The PMV index expresses a degree of comfort or discomfort for the whole body. As seen from Fig. 3(d), with an increase in the temperature difference, the PMV value is also boosted. For zero temperature difference, PMV acquired from the simulation is estimated as approximately neutral ($\text{PMV} = 0.1$) because of the 27°C operative temperature. For 12°C temperature difference, PMV predicted from the simulation exceeds $+1$, indicating to be slightly warm. It can be seen from the figures that a high radiant temperature causes an increase in the PMV, besides that, it causes local thermal discomfort especially in body parts which are close to the warm surfaces.

8. Evaluating of different building constructions commonly used in Turkey

In this section of the paper, interior surface temperatures and its effect on human thermal comfort for different wall and ceiling constructions commonly used in Turkey are analysed. The sol–air temperature approach is used to calculate the interior surface temperatures of the wall and ceiling because of its convenience. PMV index obtained from the simulation mentioned above is used to discuss thermal comfort or discomfort.

Table 3
Properties of building materials

Materials	Density, ρ (kg/m ³)	Thermal conductivity, k (W/mK)
Inner plaster	1800	0.870
Outer plaster	2000	1.400
Brick	≤ 1000	0.450
Concrete	2400	2.100
Gas concrete	800	0.290
Polystyrene foam (PS foam)	≥ 15	0.040
Extruded polystyrene foam (XPS foam)	≥ 30	0.028

Table 4
Constructions and thermal resistances of wall and ceiling

Wall/ceiling	$\sum R_i$ (m ² K/W)	$\sum R_{ti}$ (m ² K/W)
(Wall A) 2 cm inner plaster + 20 cm concrete + 3 cm outer plaster	0.26	0.14
(Wall B) 2 cm inner plaster + 19 cm brick + 3 cm outer plaster	0.59	0.47
(Wall C) 2 cm inner plaster + 20 cm gas concrete + 3 cm outer plaster	0.86	0.73
(Wall D) 2 cm inner plaster + 20 cm concrete + 4 cm XPS foam + 3 cm outer plaster	1.69	1.57
(Wall E) 2 cm inner plaster + 8.5 cm brick + 4 cm PS foam + 8.5 cm brick + 3 cm outer plaster	1.54	1.42
(Wall F) 2 cm inner plaster + 19 cm brick + 4 cm XPS foam + 3 cm outer plaster	2.02	1.90
(Ceiling A) 2 cm inner plaster + 12 cm concrete	0.20	0.08
(Ceiling B) 2 cm inner plaster + 12 cm concrete + 10 cm PS foam	2.70	2.58

In this investigation, two climatic conditions, this is the cities of Bursa and Sanliurfa in Turkey, is taken into consideration, because environmental conditions of these two cities are different from each other. Bursa is the northern-west city of Turkey and Sanliurfa is the southern-east city of Turkey. The solar radiation inputs for Bursa and Sanliurfa are calculated by the method given by Duffie and Beckman [23]. The variations of hourly ambient air temperatures and solar radiation on the horizontal surface are given in Fig. 4(a). For sol–air temperatures calculations, the term of (α/h_o) was taken as 0.052 for dark-colored surfaces [14]. The convective heat transfer coefficient, (h_o) , used for the calculations was equal to $17 \text{ W/m}^2\text{ }^\circ\text{C}$ [12–14]. The value of ΔR was taken as 63 and 0 W/m^2 for horizontal and vertical surfaces, respectively [14]. The term of ε was assumed to be 1 [14]. The variation of sol–air temperature on the horizontal surface is presented in Fig. 4(b). Also the variations of sol–air temperature and solar radiation are given for south-facing and west-facing walls in Fig. 4(c) and (d), respectively.

Different wall and ceiling constructions are taken into consideration to present the realistic building model in Turkey. Thermal properties of building materials and constructions and thermal resistances of different wall and ceiling are given in Tables 3 and 4, respectively. Six different cases depending on the wall direction and ceiling and wall types used in the investigation are presented in Table 5.

In this stage, the ambient air temperature was taken as 27°C and interior surface temperatures of the north and east walls of the room were assumed to be equal with the

Table 5
Investigated wall and ceiling constructions depend upon orientations

Case	South wall	West wall	Ceiling
1	Wall A	Wall A	Ceiling A
2	Wall B	Wall B	Ceiling A
3	Wall C	Wall C	Ceiling A
4	Wall D	Wall D	Ceiling B
5	Wall E	Wall E	Ceiling B
6	Wall F	Wall F	Ceiling B

ambient air temperature. The variations of the interior surface temperature of the south-facing wall, west-facing wall, ceiling and the variations of the mean radiant temperature and the PMV index for the city of Bursa as

a function of time for a day in June for different wall constructions are given in Fig. 5. As seen from Fig. 5, the interior surface temperatures of the un-insulated walls and ceiling such as walls (A), (B), (C) and ceiling (A) rapidly

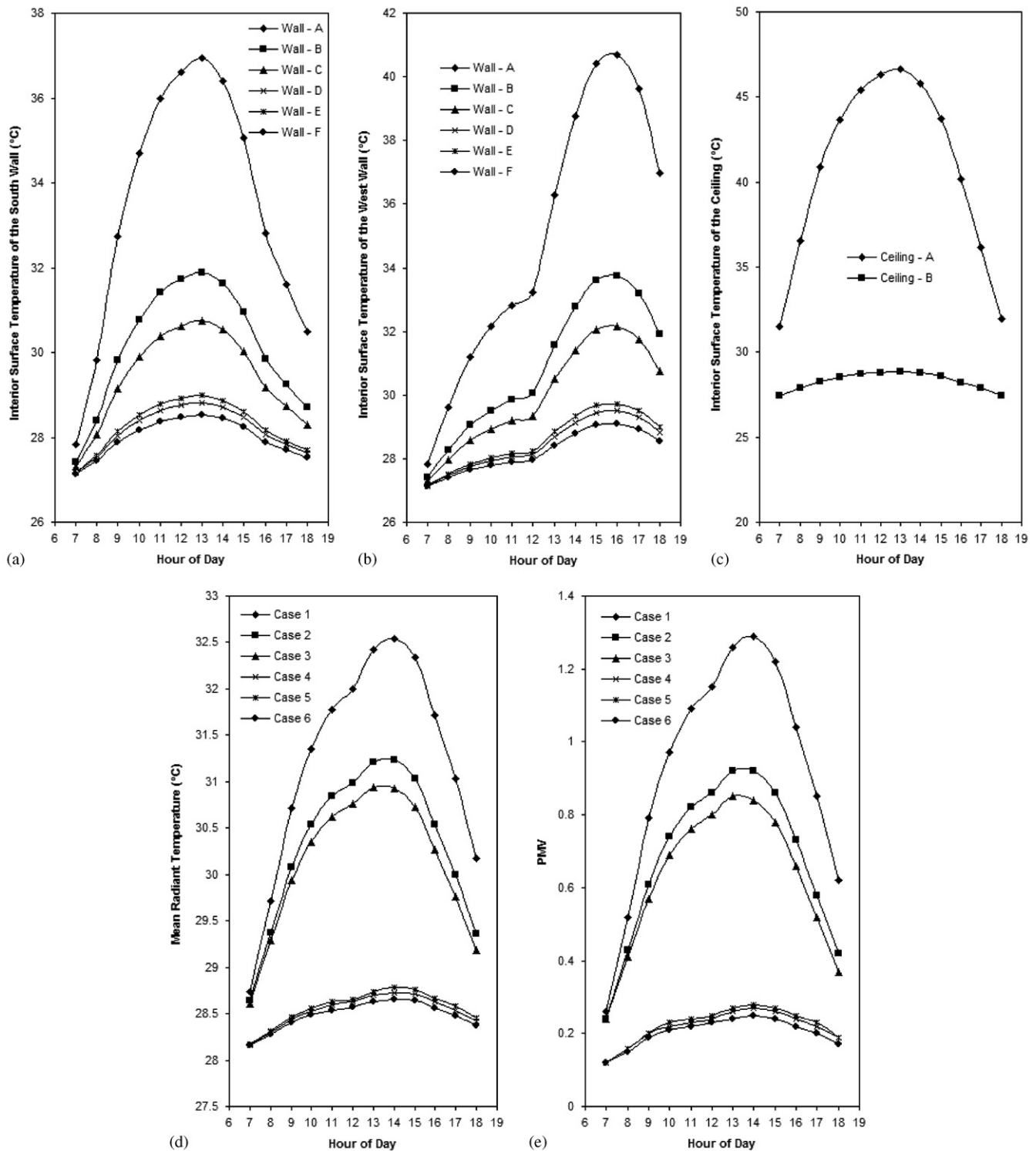


Fig. 5. (a) The variations of the interior surface temperature of the south-facing wall. (b) The variations of the interior surface temperature of the west-facing wall. (c) The variations of the interior surface temperature of the ceiling. (d) The variations of the mean radiant temperature. (e) The variations of the PMV index for the city of Bursa as a function of time for a day in June.

increase and reach high values, for instance, approximately 45°C for the ceiling and 40°C for wall (A) facing west, while the temperatures of the insulated walls and ceiling such as walls (C), (D), (E) and ceiling (B) remain approximately unchanged and remain under 30°C . It can

be seen that high solar radiation affects especially un-insulated walls. As seen from Fig. 5(d), high interior surface temperature causes high mean radiant temperature and so in this regard PMV changes in a day and reaches approximately $+1.3$ for case (1), $+0.92$ for case (2) and

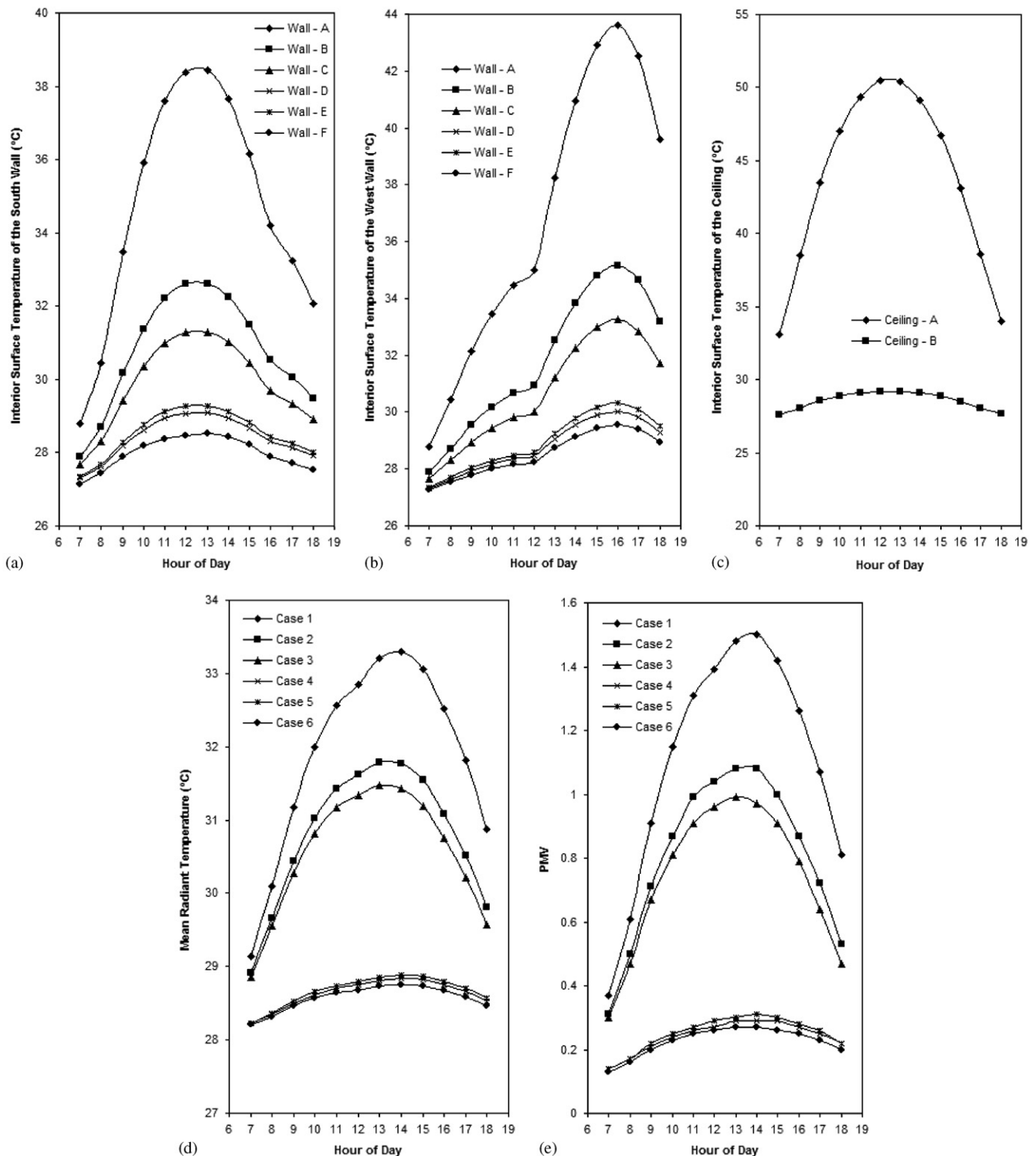


Fig. 6. (a) The variations of the interior surface temperature of the south-facing wall. (b) The variations of the interior surface temperature of the west-facing wall. (c) The variations of the interior surface temperature of the ceiling. (d) The variations of the mean radiant temperature. (e) The variations of the PMV index for the city of Sanliurfa as a function of time for a day in June.

+0.84 for case (3) indicating to be slightly warm. These values remain under +0.28 for insulated wall and ceiling constructions for cases (4)–(6).

Similar investigation is presented for the city of Sanliurfa in Fig. 6. It can be seen from Fig. 4 that the solar radiation and hourly ambient air temperatures for the city of Sanliurfa are higher than that for the city of Bursa. As shown from Fig. 6, interior surface temperatures of the un-insulated walls reach 38 °C for wall (A), 33 °C for wall (B) and 31 °C for wall (C) in the south-facing. These values are 44, 35 and 33 °C for walls (A), (B) and (C) in the west-facing, respectively. The surface temperatures of insulated walls such as walls (D), (E) and (F) remain under 30 °C. Similarly, the interior surface temperature of the un-insulated ceiling, ceiling (A), reaches 50 °C. Because of the high solar radiation and high ambient air temperature, the mean radiant temperatures for the city of Sanliurfa reach higher values than that for the city of Bursa, as seen from Fig. 6(d) and in this situation PMV index reaches approximately +1.5 for case (1), +1.1 for case (2) and +1 for case (3), indicating to be between slightly warm and warm. For the insulated walls and ceiling such as cases of (4), (5) and (6), this value remain under +0.3.

9. Conclusions

In this paper, thermal interactions between human body and its environment are simulated. This simulation is to apply the Gagge 2-node model to individual body segments rather than to whole body. The mean radiant and operative temperatures which affect body segments are determined separately, in which point the simulation is distinguished from other common thermal comfort simulations. Thus, local differences between the body segments caused by radiant temperatures are investigated. Besides this, different building constructions commonly used in Turkey are evaluated with respect to thermal comfort.

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